

Internship at AboveAir Technologies



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PAST – ACADEMIC PREPARATION

Reflecting upon my academic journey within the Applied Engineering and Technology Management (AETM) program at Millersville University, the combination of theoretical study and hands-on laboratory education has been a very key part in shaping my understanding of modern industrial systems. Entering higher education my objective was not to just learn engineering principles from a textbook, but to understand how physical components, human factors, and management strategies intersect on the factory floor. The AETM curriculum provided a structured environment where technical coursework was paired with practical applications. This allowed me to build a versatile skill set across mechanical design, electrical control systems, material processing, and operations management. This foundation showed its importance when I transitioned into the industrial workforce during my engineering internship at AboveAir Technologies, where classroom concepts immediately became my daily operational tasks.

My academic classes prepared me for manufacturing enterprise and product development. In classes like product Design I gained a deep appreciation for the lifecycle of a product. From brainstorming initial design concepts and computer-aided modeling to production scheduling and assembly line design. These courses taught me how to read and interpret complex engineering drawings, understand geometric dimensioning and tolerancing, as well as navigate structured Bills of Materials (BOMs). Understanding how a high-level assembly drawing breaks down into individual sub-assemblies and piece-part line items was essential during my internship, as it allowed me to analyze equipment schematics and identify design documentation and physical inventory on the shop floor.

Complementing the design curriculum were hands-on materials courses like Metallic and Nonmetallic Materials. Understanding the physical and mechanical properties of metals, polymers, and composites provided a small context for evaluating manufacturing processes. In these courses I learned how material selection directly impacts structural integrity, thermal performance, and ease of fabrication. Gaining experience with machining operations, sheet metal fabrication, and material joining techniques allowed me what happens on the assembly floor. When observing copper pipe bending, brazing operations, or structural cabinet fabrication in an industrial setting, my material science background allowed me to understand why specific process were necessary to prevent structural failure or fluid leaks.

One of the most technical milestones in my academic learnings occurred happened within electrical systems and power conversion coursework. In the power conversion lab we were introduced to the fundamentals of industrial automation, electrical schematics, and Programmable Logic Controllers (PLCs). I remember writing my first basic ladder logic

program to control a simple circuit where flipping an input switch would trigger an output light bulb on the lab board. While the physical outcome of lighting up a bulb was basic, the underlying logic of understanding inputs, outputs, relays, timers, and conditional statements was essential. This established a fundamental mental framework for how automated control systems operate.

Engineering classes alone won't get me anywhere without the ability to communicate concepts effectively and navigate organizational structures. Courses such as Technical Writing, Business Writing, and Principles of Management provided me with the professional communication tools necessary to bridge the gap between technical engineering teams and shop-floor personnel. Business Writing taught me how to structure concise professional correspondence, produce structured engineering reports, and present data clearly to management. Meanwhile, management coursework provided insight into workplace hierarchy, project management methodologies, and team dynamics. Additionally, my foundational courses in Precalculus and Elements of Physics provided the mathematical rigor and physical intuition needed to perform time-and-motion studies, balance line rates, and evaluate mechanical load parameters.

PRESENT – INTERNSHIP EXPERIENCE

This cumulative academic background was put to the test during the summer of 2026 when I completed the engineering internship at AboveAir Technologies. AboveAir specializes in custom designing and manufacturing, high-performance HVAC equipment for commercial and industrial applications, high standards of quality, precise assembly workflows, and strict engineering specifications. Throughout the internship I was fortunate to be mentored directly by Industrial Engineer Amy O'Neil. Amy served as an exceptional supervisor, guiding me through daily operations, teaching me how to analyze factory floor touch times, and entrusting me with real responsibility to drive lean manufacturing and continuous improvement initiatives across the facility.

My primary focus of my work at AboveAir involved observing, timing, and analyzing production workflows on active assembly lines, particularly the MC E Cabinet and Flow Line 1 production series. I conducted this as part of an ongoing effort to optimize factory throughput, I performed detailed time and motion studies by shadowing assembly technicians through complete build cycles. Using a stopwatch and tracking sheets I recorded precise touch times for every stage of fabrication. From initial structural frame assembly and sheet metal panel mounting to copper tubing installation, torch brazing, and all the way to electrical wiring. By capturing data on times and operator motion, I helped identify key line bottlenecks, non-value-added movement, and material staging delays that were slowing down overall unit production.

To help address any inefficiencies discovered during my time studies, I actively participated in weekly Kaizen continuous improvement meetings alongside production managers and design engineers. These meetings focused on eliminating waste, reducing material travel distance, and improving workstation ergonomics. I took an active role in executing 5S visual management projects across the manufacturing plant floor. To establish standardized work zones, I applied industrial floor tape throughout the plant, defining clear boundaries for Flow Line 1, Flow Line 2, raw material staging areas, and mandatory safety clearance zones around newly installed overhead cranes. Marking these walkways and work cells reduced floor clutter, improved material transport efficiency, and enhanced safety for both forklift operators and floor technicians.

Recognizing that material retrieval was a major source of wasted time for floor workers, I worked directly on material handling and inventory staging solutions. I organized copper stock storage to bring frequently used piping closer to the primary brazing stations, reducing operator walking time across the shop floor. I personally assembled and configured specialized mobile material carts, heavy-duty shelving units, and organized kanban bin storage systems for small hardware and copper fittings. By labeling and organizing these bins with clear part numbers and order parameters, floor technicians could easily locate necessary components without leaving their workstations. To further improve safety and material flow I researched and implemented methods for transport carts to prevent loose metal sheets and fittings from slipping during transport.

Among the various continuous improvement projects, my proudest accomplishment was designing and implementing an ergonomic solution for base pallet assembly. During my initial floor observations, I noticed that technicians were repeatedly bending over, kneeling, and working directly on the concrete floor when assembling shipping pallets and heavy base frames for unit transport. This practice not only presented significant ergonomic hazards and physical strain but also slowed down assembly time due to awkward working positions. I took the initiative to research industrial lifting and height-adjustable table options, evaluating weight capacities, dimensions, and floor integration. After presenting my findings to management I coordinated with Amy with and installation of an adjustable ergonomic worktable. The implementation was an immediate success as workers could now construct heavy pallet bases at a comfortable, standing working height. Eliminating constant bending directly improved worker comfort, safety, and assembly speed, demonstrating to me how targeted ergonomic investments have production benefits.

My foundational lab exercise really was noticed when I later encountered full scale commercial HVAC equipment at AboveAir Technologies and shadowed engineers

programming PLC controllers that managed multi-stage compressors, variable speed fans, and complex sensor feedback loops. It demonstrated firsthand how basic ladder logic scales directly into complex industrial automation. During the summer I had the opportunity to build off of this lab background by assisting with the programming and setup of PLC controllers installed in active HVAC units. Working directly with control systems allowed me to observe how control software logic interfaces with physical real world hardware.

My internship also provided experience in technical troubleshooting and vendor communication. Early in my internship I was tasked with diagnosing a malfunctioning Variable Frequency Drive (VFD) pump unit that suffered from unresponsive, frozen control screens. Rather than replacing parts blindly I worked through a systematic electrical troubleshooting process, evaluating power inputs and signal connections. When internal diagnostic steps did not work I contacted technical support engineers at Armstrong. The equipment manufacturer then walked through an advanced system diagnostics. After thorough troubleshooting Armstrong confirmed that the issue was an internal firmware problem and required a full replacement unit to be shipped out. This experience taught me that real-world engineering often involves coordinating with component manufacturers, managing vendor communications, and recognizing when equipment repair must yield to component replacement. In addition to mechanical assembly I contributed to workplace environmental safety and control engineering. To reduce noise pollution generated by heavy machinery in the machine shop, I researched industrial soundproofing options and contacted suppliers such as AKON to obtain formal technical quotes and material specifications for industrial sound-blocking curtains.

To support long term operational standardization at AboveAir I worked on bridging the gap between engineering documentation and employee onboarding. Recognizing that training new assembly technicians relied heavily on verbal instruction and informal shadowing, I reviewed existing physical tutorials and technical documentation to construct a comprehensive step-by-step training Power Point presentation. This training guide broke down complex cabinet assembly, electrical wiring, and component mounting into clear visual steps with labeled photos and key safety notes. Creating standardized visual work instructions ensured that new floor employees could be onboarded faster and provided the production team with a repeatable training asset.

While my internship was filled with successes, it also exposed me to the realities of floor communication breakdowns and operational friction. During one of my tasks I was instructed to label over 150 manufactured components. After spending considerable time carefully applying part identification labels across the entire inventory I was then informed

that a separate tracking label was also required for each bin. This required me to repeat the entire labeling process from scratch. While repeating the task was frustrating in the moment it showed me a lesson in manufacturing communication. It highlighted the absolute necessity of establishing clear, complete, and thorough communication prior to launching tasks to avoid unnecessary rework and wasted labor. These classes translated into my daily activities during my internship making sure I was able to communicate effectively in meetings, email, and in any projects I completed.

FUTURE – CAREER GOALS & TRAJECTORY

Looking back at my time at AboveAir Technologies has helped me realize my professional ambitions and provided a clear direction for my post-graduation career. As I complete my final semester at Millersville University targeting graduation in Fall 2026 with a B.S. in Applied Engineering & Technology Management, I feel confident in entering the industrial engineering workforce. My combination of academic coursework, practical laboratory skills, and real-world internship experience has given me a well-rounded perspective on how modern manufacturing facilities operate.

Immediately following graduation my goal is to secure an entry-level position as an Industrial, Manufacturing, or Project Engineer. While my internship provided specialized exposure to custom commercial HVAC manufacturing, I am eager to apply my skill set across a range of technical industries, including general industrial equipment fabrication, automotive manufacturing, aerospace components, or automated assembly operations. My primary criteria for an entry-level role is to be in an environment that allows me to remain hands-on with physical processes, stay active on the production floor, and actively work on optimizing manufacturing workflows, eliminating operational waste, and integrating smart automation.

To complement my academic degree and position myself for rapid career growth, I am actively pursuing key professional credentials. I currently hold an OSHA 10-Hour General Industry Certification, which has ingrained a safety first mindset across all floor activities. To enhance my capability in data analytics and time-study formatting, I am currently working toward earning an Advanced Excel Certification that will allow me to perform advanced statistical analysis, build automated data models, and create clear operational dashboards. Within 3 to 5 years post-graduation I also plan to achieve a Lean Six Sigma Green Belt certification to further formalize my continuous improvement and statistical process control methodology.

In the long term my ultimate career goal is to transition from individual technical engineering roles into managerial leadership as an Engineering Manager. I believe that

effective engineering leadership requires a first-hand understanding of the work performed by floor technicians combined with a strategic understanding of facility operations, safety compliance, and business objectives. By starting my career on the manufacturing floor, solving daily production bottlenecks, and working alongside cross functional teams, I am building the foundational experience necessary to lead technical personnel, manage multi-million dollar manufacturing facilities, and drive operational excellence. My journey through Millersville University and AboveAir Technologies has laid the groundwork for this career trajectory, instilling in me the technical competence, practical problem-solving ability, and leadership drive needed to succeed in the modern industrial landscape.